11 Many-period problem

We did a two period model, fine. What about many periods? It turns out a many-period model isn't very different from a one-period model, just like a many-good model isn't too different from a two-good model in a single period of time. The only difficulty is we often calculate infinite-period models, which adds a wrinkle in the things we have to do.

Take an endowment economy with a constant endowment in each period, and a constant interest rate. We do the same thing we did before: get the net present value of income. We start with:

$$\begin{aligned}
c_1 + s_1 &= I_1 \\
c_2 + s_2 &= I_2 + (1+r)s_1 \\
c_2 + s_2 &= I_2 + (1+r)s_1 \\
\vdots \quad & \quad \vdots \\
c_n + s_n &= I_n + (1+r)s_{n-1} \\
\vdots \quad & \quad \vdots
\end{aligned}$$

All the way out to infinity. When we start sewing these together by plugging in s's (as we did in the two-period model) we get the net present value budget constraint:

$$\sum_{t=1}^{\infty} \frac{c_t}{(1+r)^{t-1}} = \sum_{t=1}^{\infty} \frac{I_t}{(1+r)^{t-1}}$$

Recall that we can flip this to something more comfortable, because:

$$\sum_{k=0}^{\infty} \frac{c}{(1-r)^k} = \frac{c}{1-\frac{1}{1-r}} = c\frac{1-r}{r}$$

Therefore, changing the summation index and applying the formula:

$$\sum_{t=1}^{\infty} \frac{c_t}{(1+r)^{t-1}} = \frac{1-r}{r} I$$

So we have the Lagrangian:

$$L = \sum_{t=1}^{\infty} \beta^{t-1} \log(c_t) + \lambda \left(\frac{1-r}{r} I - \sum_{t=1}^{\infty} \frac{c_t}{(1+r)^{t-1}} \right)$$

Taking generic first order condition of c_t:

$$\frac{\partial L}{\partial c_t} : \beta^{t-1} \frac{1}{c_t} = \lambda \frac{1}{(1+r)^{t-1}}$$

Or, perhaps more nicely:

$$((1+r)\beta)^{t-1} \frac{1}{c_t} = \lambda$$

This will hold for *all* $t \in \{1, 2, ...\}$.
In both the labor market and the two-period model, we've stopped and taken a breath and analyzed the Lagrange multiplier. No sense in stopping now. The Lagrange multiplier is how much utility we get when we relax the budget constraint by one unit (in this case, a unit of endowment, of consumption). What this sent of Euler equations is saying is that all our consumption bundles are linked over time. For instance, if we have the case that $\beta = \frac{1}{1+r}$, then:

$$\frac{1}{c_1} = \frac{1}{c_2} = ... = \frac{1}{c_n} = \lambda$$

Consumption never grows, it's always the same. If, instead, $\beta(1+r) = 0.99$, then:

$$c_1 = 0.99c_2 = 0.99^2c_3 = ... = 0.99^nc_n = \lambda$$

We have all the consumptions relative to one another: that's what the Euler equation's given us. The budget constraint tells us what they all have to add up to, in net present value terms. In this case, if c_1 is 1, then $c_2 = 1.\overline{01}$, and $c_3 = 1.023$: the interest rate is higher than our impatience, and we'll keep saving forever, getting higher and higher consumptions as we go.